# johnny_4_docs

System-level documentation for the Johnny 4 robot: everything that spans more than one board repo.

## Contents

| File | What it is |
|---|---|
| POWER.md | Power distribution for the robot side: which pin each board takes 5V on, buck converter zones, USB flashing/backfeed rules, budget |
| j4_master_pin_diagram.pdf | All seven boards' pin diagrams, one full letter page each, printable as one document |
| tools/gen_master_pin_pdf.py | Regenerates the master PDF (and the standalone per-repo pin diagrams you name) from the board repos' READMEs |

## Regenerating the master pin diagram

The generator parses each repo README's `## Pin diagram` fenced block and `## Pin assignments` table, so the PDF always matches the repos. After changing any pinout, update that repo's README and run:

```
python tools/gen_master_pin_pdf.py
```

That always rewrites `j4_controller_pin_diagram.pdf` too. To rewrite other boards' standalone `<repo>_pin_diagram.pdf` from the same pages, name them:

```
python tools/gen_master_pin_pdf.py j4_display_right j4_talk
```

Only the boards named are touched.

Requires `reportlab`. The script expects the board repos checked out as siblings of this repo ( `../j4_controller`, `../j4_receiver`, and so on).

## The board repos

Signal chain: j4_controller --ESP-NOW--> j4_receiver --UART--> j4_stepper_neck --UART--> j4_stepper_eyes, with j4_talk on the receiver's second UART and j4_display_left + j4_display_right on the controller's UARTs. Printable parts live in johnny_4_3D_printable_parts.